GATE

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GENERAL APTITUDE

GQ

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General Aptitude
GQ on Quantitative Aptitude

Quantitative:

1. Simplify: $\dfrac{18^{3/2} \times 12^{-1/2} \times 15^{-1/2} \times 8^{4/3}}{27^{2/3} \times 10^{-1/2} \times 81^{3/4}}$

[1] $\dfrac{27}{32}$ **[2]** $\dfrac{32}{64}$ **[3]** $\dfrac{32}{27}$

[4] $\dfrac{1}{3}$ **[5]** None of these

2. Simplify:

$$\left[\frac{\left\{1+\overline{8-\frac{4}{5}} \quad \text{of} \quad 3\frac{2}{9}\right\} \div \frac{11}{25} + 41}{\left(81^{1/4}\right)^3 \div \overline{3\frac{1}{2} + 1} - (1728)^{\frac{1}{3}}}\right]$$

[1] -19 **[2]** -16 **[3]** -24
[4] 0 **[5]** None of these

3. Three years ago, the average age of Sita, Gita and Mita was 27 years and 5 years ago average age of Gita and Mita was 20 years. What is Sita's present age?
[1] 32 years **[2]** 36 years **[3]** 40 years
[4] 44 years **[5]** Cannot be determined

4. By scoring 100 in his 30th innings, Dhoni increased his average by 1.5. What was his average for the first 29 innings?
[1] 60 **[2]** 55 **[3]** 40
[4] 50 **[5]** None of these

5. What is the 99th term of the series 2 + 7 + 14 + 23 + 34 +… ?
[1] 10000 **[2]** 9997 **[3]** 9999
[4] 9998 **[5]** None of these

Set Theory:

6. A dinner party is to be fixed for a group consisting of 100 persons. In this party, 50 persons do not prefer fish, 60 prefer chicken and 10 do not prefer either chicken or fish. What is the number of persons who prefer both fish and chicken?
[1] 15 **[2]** 25 **[3]** 10
[4] 20 **[5]** 30

7. In a group of 300 students who have opted for either MMS or PGDBM, 64 % students are opted for MMS and 52 % for PGDBM. If the students opted for both finally select MMS vs PGDBM in the ratio 1 : 3 , then how many more students have opted for MMS than for PGDBM.
[1] 20 **[2]** 4 **[3]** 12
[4] 16 **[5]** None of these

Directions (Qs.8 to 10): Answer the questions based on the following information.

In a survey of students of JBIMS, it was found that 58 % students like Bourn vita, 42 % students like Horlicks, 40 % students like Complan, 22 % like only Bourn vita and Horlicks, 18 % liked only Bourn vita and Complan, 14 % like only Horlicks and Complan and 6 % like all the three drinks.

8. What is the ratio of number of students who liked only Bourn vita to that of students who liked only Complan?
[1] 3 : 1 **[2]** 6 : 1 **[3]** 12 : 1
[4] 1 : 1 **[5]** 2 : 1

9. What percent of students do not like any of the three drinks?
[1] 76 % **[2]** 24 % **[3]** 6 %
[4] 94 % **[5]** None of these

10. What is the percentage of the students who like at the most two drinks?
[1] 90% **[2]** 76% **[3]** 24%
[4] 94% **[5]** None of these

11. How many 5 digit numbers can be formed from the digits 2, 0, 5, 6, 9, 3 and 7 if the repetition of digits is not allowed?
[1] 2520 **[2]** 2160 **[3]** 720
[4] 5040 **[5]** None of these

12. In how many ways can a group of 5 girls and 2 boys be made out of 7 girls and 3 boys?
[1] 81 **[2]** 72 **[3]** 69
[4] 63 **[5]** 126

13. There are 11 pairs of socks of different sizes. In how many ways can you choose one for the left foot and one for the right foot such that they are not of the same pair?
[1] 121 **[2]** 132 **[3]** 110
[4] 120 **[5]** None of these

14. In a simultaneous throw of two dice, what is the probability of getting a total of 10 or 11?
[1] $\frac{1}{2}$ **[2]** $\frac{5}{18}$ **[3]** $\frac{7}{36}$
[4] $\frac{2}{12}$ **[5]** $\frac{5}{36}$

15. A box has 10 oranges, out of which 4 are rotten. If 3 oranges are chosen at random, then find the probability that none of them is rotten?
[1] $\frac{2}{9}$ **[2]** $\frac{1}{6}$ **[3]** $\frac{4}{9}$
[4] $\frac{1}{7}$ **[5]** $\frac{5}{4}$

16. A bag contains 5 yellow and 4 pink balls. Three balls are drawn at random. What is the probability that the two are yellow and one is pink?

[1] $\frac{1}{2}$ **[2]** $\frac{1}{6}$ **[3]** $\frac{1}{3}$

[4] $\frac{1}{4}$ **[5]** None of these

17. If P's weight is 15 % less than Q's weight then by what percentage Q's weight is more than P's weight?

[1] $17\frac{17}{11}\%$ **[2]** 20% **[3]** $15\frac{11}{17}\%$

[4] 18% **[5]** $17\frac{11}{17}\%$

18. If the length of a rectangle is increased by 25% and its breadth is decreased by 10%, then find the percentage change in the area of the rectangle?

[1] 1% **[2]** 40% **[3]** –12.5%
[4] 10% **[5]** 12.5%

19. Due to reduction in the price of apples by 30%, Ravina can buy 15 apples more in the same amount. Find the original number of apples.

[1] 50 **[2]** 40 **[3]** 20
[4] 35 **[5]** 45

20. In an examination, Bhakti scored 25% of total marks and failed by 100 marks whereas Sandhya scored 45% of total marks and she got 40 marks above pass marks. Find out the total marks for the examination.

[1] 750 **[2]** 700 **[3]** 800
[4] 600 **[5]** None of these

21. James sold a shirt at a gain of 10 %. If he had bought the shirt at 10 % less and sold it for Rs.132 less, he would have still gained 10 %. What is the CP of the shirt?

[1] Rs.900 **[2]** Rs.1000 **[3]** Rs.1200
[4] Rs.1100 **[5]** Rs.1500

22. At what per cent above the cost price must a watch be marked so as to gain 4 % after allowing a discount of 20 %?

[1] 24 % **[2]** 30 % **[3]** 28 %
[4] 32 % **[5]** 40 %

23. Maganlal buys 2 varieties of rice, Basmati and Kovalam. He bought Kovalam at Rs.30 per kg and Basmati at Rs.50 per kg. He sold all the rice at Rs.40 per kg. What is the profit or loss percentage of Maganlal if he spent an equal amount of money on both the varieties of rice?

[1] No loss, no profit **[2]** 5 % loss **[3]** 6.67 % gain
[4] 5 % profit **[5]** 6.67 % loss

24. Aamir travels at the rate of 3 kmph and he reaches 15 minutes late to his studio. If he travels at the rate of 4 kmph from his house, he reaches his studio 15 minutes earlier. What is the distance of Aamir's house from his studio?
[1] 9 km **[2]** 12 km **[3]** 0 km
[4] 8 km **[5]** 6 km

25. A jet plane covers first 1600 km with the speed of 400 km/hr, next 1200 km with the speed of 300 km/hr and the last 500 km with the speed of 250 km/hr. What is the average speed of the jet plane for the entire journey?
[1] 300 km/hr **[2]** 316.67 km/hr **[3]** 275 km/hr
[4] 285 km/hr **[5]** 330 km/hr

26. Charles steals a car at 11 a.m. and drives it at 45 km/hr. The theft is discovered at 12.30 p.m. and the police starts to chase him at the speed of 60 km/hr. At what time the Charles will be caught if police chases him in the right direction?
[1] 5.30 p.m.
[2] 4.30 p.m.
[3] 4.45 p.m.
[4] 5 p.m.
[5] Charles will never be caught

27. D can dig a well of particular size in 8 days while E can fill up the same well in 3 days. D has worked for 4 days and then E joined him for another 2 days only. In how many days will D alone dig up the remaining part of the well?
[1] 7 days **[2]** $13\frac{1}{3}$ days **[3]** $7\frac{1}{3}$ days
[4] $6\frac{1}{3}$ days **[5]** None of these

28. In a factory, 15 men can fill 35 boxes in 7 hrs. How many men can fill 65 boxes in 5 hrs?
[1] 26 **[2]** 39 **[3]** 42
[4] 60 **[5]** 45

29. If 7 men or 5 boys or 6 women take 12 days to do a job, then how much time would they take to complete the same job working together?
[1] 2 days
[2] $5\frac{2}{3}$ days
[3] 4 days
[4] $3\frac{1}{8}$ days
[5] None of these

Simple Table:

The following table shows the domestic sales (in thousands) of scooters by five manufacturers from 2001 to 2006.

Manufacturer	2001	2002	2003	2004	2005	2006
Bajaj	44	49	47	55	60	64
Kinetic	40	55	50	63	72	80
LML	38	45	56	60	65	71
Honda	36	48	60	72	84	81
Tata	45	40	48	57	66	75

30. Considering the period given, the domestic sales of which manufacturer is the highest?
[1] Bajaj **[2]** Tata **[3]** Kinetic
[4] LML **[5]** Honda

31. What is the difference between the average sales of scooters per year by Kinetic and the average sales of scooters per year by Honda in the given period?
[1] 3000 **[2]** 3500 **[3]** 3300
[4] 3100 **[5]** 3700

32. What is the growth rate of LML for the year 2005?
[1] 5 % **[2]** 6.67 % **[3]** 10 %
[4] 8.33 % **[5]** None of these

33. What is the growth rate of Honda for the year 2001?
[1] 15 % **[2]** 10 % **[3]** 12.5 %
[4] 4 % **[5]** Data Insufficient

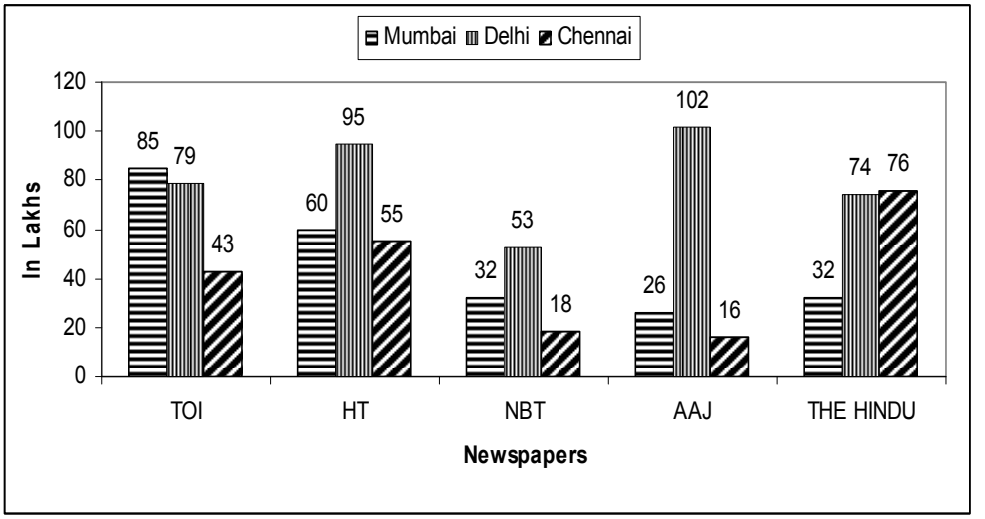
34. During 2004, the sales of which manufacturer has shown the maximum percentage increase over the previous year?
[1] Tata **[2]** LML **[3]** Kinetic
[4] Bajaj **[5]** Honda

35. What is the simple annual growth rate of sales of Kinetic scooters from 2001 to 2006?
[1] 20 % **[2]** 100 % **[3]** 25 %
[4] 10 % **[5]** None of these

36. What is the compounded annual growth rate of sales of Tata scooters from 2001 to 2006?
[1] 40 % **[2]** 66.67 % **[3]** 25 %
[4] 11 % **[5]** 5%

Simple Bar Graph:

The graph below gives the distribution of five different newspapers over three different cities. The figures are in lakhs of copies.



37. Which of the following newspapers has the highest sales for all the given cities taken together?
[1] TOI **[2]** HT **[3]** NBT
[4] AAJ **[5]** THE HINDU

38. Sale of which of the following is the least for the three cities put together?
[1] TOI **[2]** HT **[3]** NBT
[4] AAJ **[5]** THE HINDU

39. What is the ratio of the maximum sales to the minimum sales of any newspapers for any city?
[1] 7 : 3 **[2]** 17 : 3 **[3]** 51 : 13
[4] 51 : 8 **[5]** None of these

40. By how much percentage the sales of THE HINDU in Delhi is more than the sales of the NBT in Mumbai?
[1] 131.25 % **[2]** 89.75 % **[3]** 152.5 %
[4] 100 % **[5]** 203.35 %

41. By how much the combined sales of HT and AAJ is more than the combined sales of NBT and TOI for all the three given cities?
[1] 4.4 lakhs **[2]** 26 lakhs **[3]** 350 thousands
[4] 47 lakhs **[5]** 44 lakhs

Reasoning:

Directions (Qs. 42 to 44): The cast of a benefit program will consist of exactly four actors to be selected from the seven actors P, Q, R, S, T, U and V according to the following.

conditions:

(i) Either P or T must be in the cast; but they cannot both be in the cast.
(ii) If T is in the cast, then Q must be in the cast,
(iii) If U is in the cast, then neither P nor R can be in the cast?

42. If U is in the cast, then which of the following must also be in the cast?
[1] P **[2]** Q **[3]** R
[4] S **[5]** V

43. If P is in the cast, then any of the following groups of three could make up the rest of the cast except
[1] Q, R and S **[2]** Q, R and V **[3]** R, S and V
[4] S, U and V **[5]** None of these

44. If S is in the cast but needs to be replaced, then which of the following actors, if not in the cast, is eligible to replace S regardless of the composition of the rest of the cast?
[1] R **[2]** P **[3]** T
[4] V **[5]** U

Directions (Qs. 45 to 47): Study the following information carefully and answer following questions:
(i) A, B, C, D, E and F are six members of a family,
(ii) There are two married couples among them.
(iii) C is the mother of A and F.
(iv) E is the father of D.
(v) A is the grandson of B.
(vi) The total number of female members in the family is three.

45. How is B related to F?
[1] Sister **[2]** Grandmother **[3]** Wife
[4] Daughter **[5]** Aunt

46. How is F related to A?
[1] Sister **[2]** Brother **[3]** Daughter
[4] Mother **[5]** None of these

47. Which of the following is a group of female members?
[1] BDE **[2]** AFB **[3]** DCF
[4] BCF **[5]** None of these

Directions (Qs. 48 to 52): Study the following information carefully to answer these questions.

M, V, K, P, T, W and H are seven friends studied in three management institutes A, B and C. They had opted for one out of the three specializations Marketing, Finance and Personnel with at least two of them in each specialization. The Marketing specialist from

institute B earns the maximum. V studied in institute C with Personnel specialization and earns more than only M and P. K studied in institute A with Marketing and earns less than only T and H. W studied in institute C with Finance and earns more than V but less than K. H studied in institute A with Finance specialization. P, a Personnel specialist studied in institute B and is not the least earner among them. M is the only Finance specialist from one of these institutes among them.

48. What is T's specialization?
[1] Marketing **[2]** Finance **[3]** Personnel
[4] Data inadequate **[5]** None of these

49. Whose income is the second highest among them ?
[1] T **[2]** K **[3]** H
[4] T or H **[5]** V

50. How many of them earn more than V?
[1] Two **[2]** Three **[3]** Five
[4] One **[5]** None of these

51. Which of the following combinations of institute-individual-specialization is not correct?
[1] A—K—Marketing **[2]** B—P—Personnel **[3]** C—W— Finance
[4] A—P—Personnel **[5]** B—M—Finance

52. In which of the institutes did M study ?
[1] A **[2]** B **[3]** C
[4] B or C **[5]** Data inadequate

Directions (Qs.53 & 54): Questions are based on a code language. Study the relationship and pick one of the options that best corresponds to the given relationship.

53. If the word LOVE is coded as 27, then what should be the code for word HATE?
[1] 27 **[2]** 34 **[3]** 72
[4] 15 **[5]** 17

54. If 'CET' stands for 300, and 'CAT' stands for 60, then what does 'MMS' stand for?
[1] 500 **[2]** 3211 **[3]** 4711
[4] 2971 **[5]** None of these

Directions (Qs.55 & 56): In the questions given below, if the given interchanges are made in signs and numbers, which one of the five equations would be correct.

55. Given interchanges: 'signs $\div$ and $\times$' ; 'numbers 8 and 4'.
[1] $4 \times 8 \div 2 = 5$ **[2]** $4 \div 8 \times 8 = 7$ **[3]** $4 \times 8 \div 4 = 16$
[4] $4 \div 8 \times 2 = 17$ **[5]** $8 \times 2 \div 8 = 2$

56. Given interchanges: 'signs + and –' ; 'numbers 5 and 3'.
[1] $3 - 5 + 2 = 7$ **[2]** $3 + 5 - 4 = 7$ **[3]** $3 - 5 - 2 = 11$
[4] $5 - 3 \div 2 = 1$ **[5]** $3 + 5 - 6 = 8$

57. If the first and third, second and fourth, fifth and seventh, sixth and eight letter of the word 'REVISION' is interchanged, then which is the 3rd letter to the right of 'E'?
[1] V **[2]** N **[3]** R
[4] S **[5]** O

58. If the word 'LAPTOP' is coded using the reverse series logic (for example in reverse series logic, letter 'A' will be coded as 'Z', letter 'B' will be coded as 'Y' and so on). What will be the fourth letter from the left in the code?
[1] G **[2]** Z **[3]** K
[4] L **[5]** O

Directions (Qs.59 & 60): In each of the following questions, certain code values are assigned to a word and you have to find out the code which will be used for the given word, from the given options.

59. If DOGRA is written as GODAR, then how DREAM will be written in this code?
[1] DERMA **[2]** ERDMA **[3]** ERMDA
[4] EDRMA **[5]** None of these.

60. If in a certain code SOCIAL is written as TPDJBM, then how BICYCLE will be written in this code?
[1] CJDZEMF **[2]** CKDXDMF **[3]** CJDZDMF
[4] CJDZDKG **[5]** None of these.

61. Jaya faces north and covers 24 km, turns west and covers 12 km, then turns south and covers 6 km and then turns west again and covers 12 km. How far is she from the starting point and in which direction?
[1] 20 km north-east **[2]** 25 km South-west **[3]** 40 km north-west
[4] 30 km north-west **[5]** None of these

62. If RAMAN is written as 12325 and DINESH as 675894, then how will MANDAR be written?
[1] 325621 **[2]** 325126 **[3]** 325612
[4] 325261 **[5]** None of these

63. Find the number which will come in place of the question mark.

7A	9C	2B
14B	6A	?
9C	15B	23A

[1] 7C **[2]** 5C **[3]** 9C
[4] 10C **[5]** 8C

64. At a parking lot, 36 vehicles are parked in a single row. After the first tractor there is one bicycle, after the second tractor there are two bicycles after the third tractor there are three bicycles, and so on. What is the number of bicycles in the row?
[1] 35 **[2]** 28 **[3]** 26
[4] 21 **[5]** None of these

Directions (Qs. 65 to 68): Five ambassadors assemble in Geneva for a conference.

P can speak in Spanish and Hindi.
Q speaks Spanish and English.
R converses in English and Hindi.
S speaks French and understands Spanish.
T can also speak French and Hindi.

65. Which of the following cannot converse without an interpreter?
[1] Q and T **[2]** P and Q **[3]** P and R
[4] Q and S **[5]** P and T

66. If a sixth ambassador is brought in, to be understood by the maximum number of the original five, he should be fluent in
[1] English and French **[2]** Hindi and English **[3]** Hindi and Spanish
[4] English and Spanish **[5]** French and Hindi

67. Which of the following can act as an interpreter when R and S wish to talk?
[1] Only P **[2]** Only Q **[3]** Only T
[4] P or Q **[5]** Any of the three ambassadors

68. Besides T, which of the following can converse with S without an interpreter?
[1] Only P **[2]** Only Q **[3]** Only R
[4] P and Q **[5]** P, Q and R

Directions (Qs.69 to 72): For the following questions.
Mark: **[1]** if x is greater than y.
[2] if x is less than y.
[3] if x is equal to y.
[4] if x is greater than or equal to y.
[5] if x is less than or equal to y.

69. (I) $x^2 = 64$ (II) $y = 8$

70. (I) x = square root of 169 (II) $y^2 = 169$

71. (I) $x^2 - 7x + 12 = 0$ (II) $y^2 + 5y + 6 = 0$

72. (I) x is the largest angle in a right angled triangle
(II) y is the angle between the diagonals of a rhombus.

Directions (Qs. 73 to 76): Find the next term in the series.

73. 1, 3, 7, 15, 31, ---
[1] 60 **[2]** 64 **[3]** 63
[4] 75 **[5]** 57

74. 4, 7, 10, 11, 22, 17, 46, 25, ---
[1] 28 **[2]** 94 **[3]** 96
[4] 32 **[5]** 102

75. 12, 43, 95, 167, ---
[1] 369 **[2]** 3611 **[3]** 2511
[4] 259 **[5]** None of these

76. 17, 17, 68, 612, ---
[1] 10000 **[2]** 9764 **[3]** 9842
[4] 9542 **[5]** 9792

Directions (Qs. 77 to 79): Find the wrong term in the series.

77. 1, 3, 5, 7, 11, 13,…
[1] 1 **[2]** 3 **[3]** 5
[4] 11 **[5]** 13

78. 555, 557, 554, 559, 552, 561, 550, 567
[1] 559 **[2]** 550 **[3]** 561
[4] 567 **[5]** 552

79. 0, 0, 0, 1, 1, 2, 2, 4, 8, 3, 9, 27
[1] 1 **[2]** 4 **[3]** 2
[4] 0 **[5]** 27

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ANSWER KEY

1.	**[3]**	2.	**[2]**	3.	**[3]**	4.	**[2]**
5.	**[4]**	6.	**[4]**	7.	**[3]**	8.	**[2]**
9.	**[5]**	10.	**[4]**	11.	**[2]**	12.	**[4]**
13.	**[3]**	14.	**[5]**	15.	**[2]**	16.	**[5]**
17.	**[5]**	18.	**[5]**	19.	**[4]**	20.	**[2]**
21.	**[3]**	22.	**[2]**	23.	**[3]**	24.	**[5]**
25.	**[5]**	26.	**[4]**	27.	**[3]**	28.	**[2]**
29.	**[3]**	30.	**[5]**	31.	**[2]**	32.	**[4]**
33.	**[5]**	34.	**[3]**	35.	**[1]**	36.	**[2]**
37.	**[2]**	38.	**[3]**	39.	**[4]**	40.	**[1]**
41.	**[5]**	42.	**[2]**	43.	**[4]**	44.	**[4]**
45.	**[2]**	46.	**[1]**	47.	**[4]**	48.	**[1]**
49.	**[3]**	50.	**[5]**	51.	**[4]**	52.	**[2]**
53.	**[5]**	54.	**[2]**	55.	**[3]**	56.	**[5]**
57.	**[4]**	58.	**[1]**	59.	**[2]**	60.	**[3]**
61.	**[1]**	62.	**[1]**	63.	**[2]**	64.	**[2]**
65.	**[1]**	66.	**[3]**	67.	**[5]**	68.	**[4]**
69.	**[5]**	70.	**[4]**	71.	**[1]**	72.	**[3]**
73.	**[3]**	74.	**[2]**	75.	**[3]**	76.	**[5]**
77.	**[1]**	78.	**[3]**	79.	**[3]**		

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